

VLSI DESIGN INTERNSHIP

Task 1 — Technical Report

Overview of the VLSI Design Process and Simulation of Fundamental Digital Logic Circuits

Simulation Platform: Logisim

Domain: Digital Design & VLSI

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Executive Summary

This report documents Task 1 of a VLSI design internship, focused on building a working understanding of the VLSI design flow and validating the behaviour of fundamental digital logic circuits. Seven basic logic gates — NOT, AND, OR, NAND, NOR, XOR, and XNOR — were implemented and functionally verified in Logisim, followed by the design and verification of a Half Adder circuit. All outputs were cross-checked against their expected truth tables and matched as predicted, confirming correct circuit behaviour.

Introduction to the VLSI Design Flow

Very Large-Scale Integration (VLSI) technology enables millions of transistors to be integrated onto a single semiconductor chip. It forms the foundation of modern electronic systems, including microprocessors, memory modules, communication devices, and everyday consumer electronics.

A standard VLSI design methodology follows a structured sequence of stages, taking a design from concept through to a fabricated and validated chip:

Standard Design Methodology

Requirement Analysis → System Architecture → RTL Coding → Functional Verification → Logic Synthesis → Physical Design → Sign-off Verification → Fabrication → Testing and Validation

This task focuses on the foundational stage of this flow: verifying combinational logic behaviour through simulation, which underpins all higher-level RTL and synthesis work later in the pipeline.

Digital Logic Gate Simulations

The behaviour of seven essential digital logic gates was implemented and verified using Logisim, a circuit simulation tool widely used for digital logic education and prototyping. Each gate was constructed with input pins and probes, exercised across all input combinations, and its output logic level was recorded.

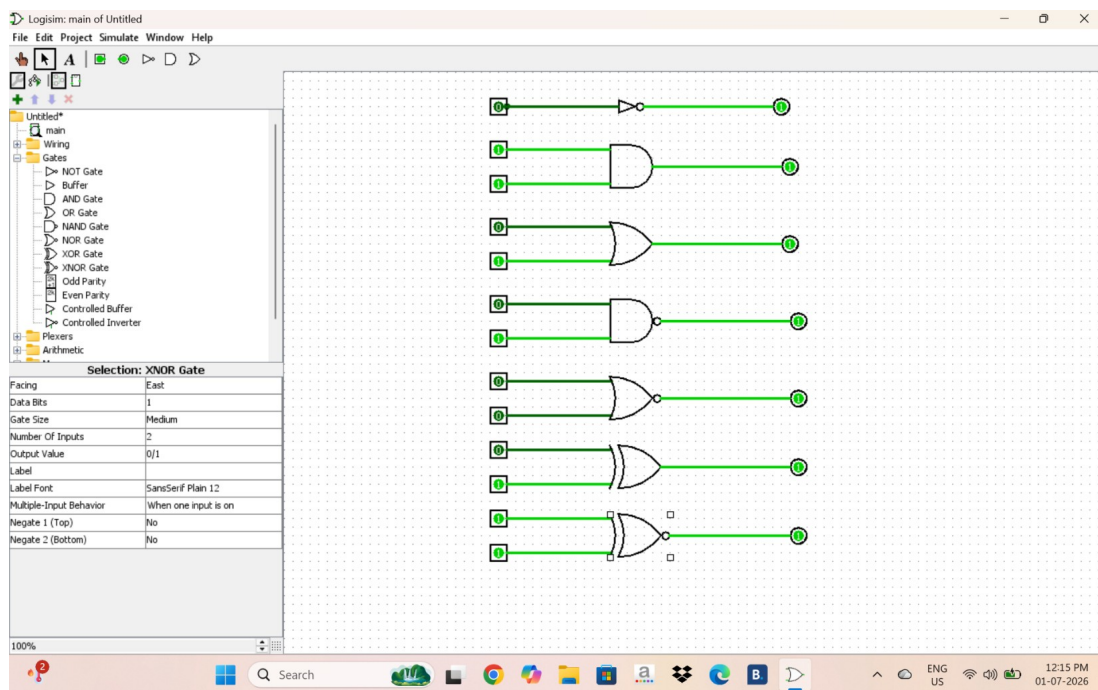


Figure 1. Logisim implementation of the seven fundamental logic gates.

Each gate — NOT, AND, OR, NAND, NOR, XOR, and XNOR — was built independently using Logisim's gate library, with input pins driven manually and output probes used to observe the resulting logic levels. This confirmed correct wiring and functional behaviour for every gate prior to truth-table verification.

Truth Table Verification

The output of each gate was recorded for every possible input combination and compared against the expected Boolean truth table. All results matched the expected behaviour, confirming correct gate-level implementation.

NOT Gate

$$Y = A'$$

A	Y
0	1
1	0

AND Gate

$$Y = A \cdot B$$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate

$$Y = A + B$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NAND Gate

$$Y = (A \cdot B)'$$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate

$$Y = (A + B)'$$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

XOR Gate

$$Y = A \oplus B$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

XNOR Gate

$$Y = (A \oplus B)'$$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Half Adder Design and Implementation

Building on the verified logic gates, a Half Adder circuit was designed and simulated. The Half Adder is a fundamental combinational arithmetic circuit that adds two single-bit binary inputs and produces a Sum and a Carry output, forming a basic building block for larger arithmetic circuits such as Full Adders and multi-bit adders.

Logic Equations

$$\text{Sum (S)} = A \oplus B \quad \text{Carry (C)} = A \cdot B$$

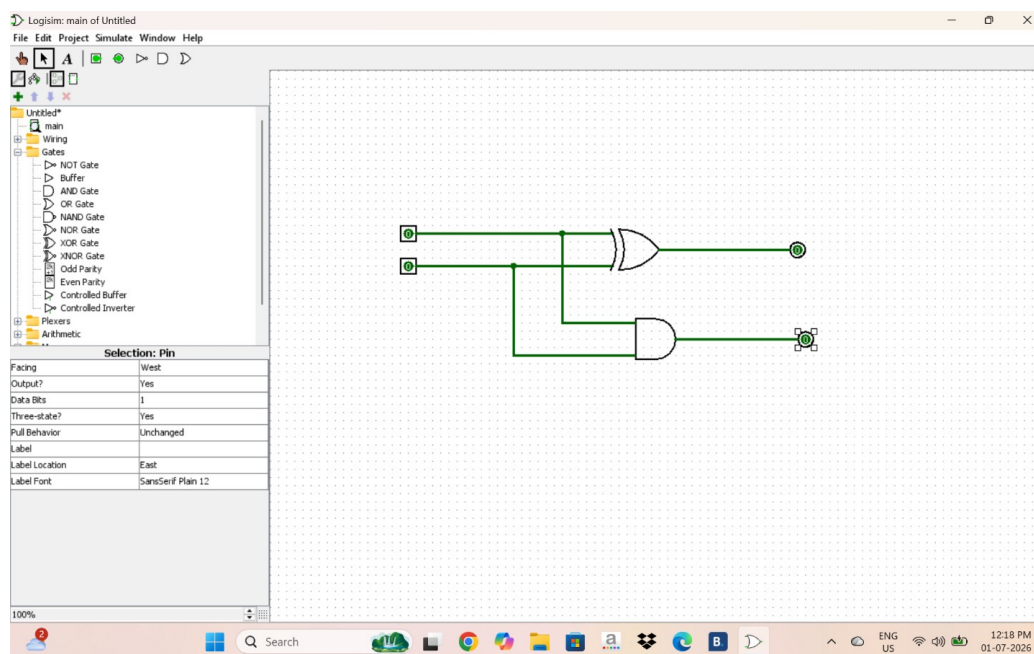


Figure 2. Logisim implementation of the Half Adder circuit.

The circuit was built using an XOR gate to generate the Sum output and an AND gate to generate the Carry output, both driven by the same two input pins. Simulation across all input combinations confirmed the Sum and Carry outputs matched the expected arithmetic behaviour.

Half Adder Truth Table

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Observations

- All seven logic gates produced outputs that matched their expected truth tables across every input combination.
- The XOR gate correctly produced a high output only when its two inputs were unequal, confirming its distinguishing behaviour relative to OR.
- The Half Adder circuit correctly generated the Sum using an XOR gate and the Carry using an AND gate, matching the expected arithmetic truth table.
- Simulation in Logisim provided a clear, visual method for verifying combinational logic behaviour before moving to RTL-level design.

Key Learnings

- Gained a working understanding of the end-to-end VLSI design methodology, from requirement analysis through to fabrication and testing.
- Reinforced the Boolean logic and truth-table behaviour of the seven fundamental logic gates through hands-on simulation.
- Understood how basic gates combine to form arithmetic building blocks, using the Half Adder as a practical example.
- Practised using Logisim as a simulation tool for constructing, testing, and debugging digital circuits.

Conclusion

Task 1 was successfully completed using Logisim simulations. All seven fundamental logic gates and the Half Adder circuit were implemented, tested, and verified against their expected truth tables, with all results matching expected behaviour. This work establishes the foundational digital logic understanding required for subsequent, more advanced tasks in the VLSI design flow.